
Do Spatial Multi-Head Self-Attention Modules Improve VGG-Style ConvNets on CIFAR-10?

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Abstract

This project studies whether spatial multi-head self-attention improves a strong VGG-style convolutional network trained from scratch on CIFAR-10. The basic project first upgrades an Assignment 3 patchify ConvNet into a modern baseline using stacked VGG-style convolutional blocks, batch normalization, data augmentation, AdamW weight decay, warmup followed by cosine learning-rate decay, and global average pooling. The B/A-extension then inserts transformer-style spatial self-attention modules after selected convolutional blocks and evaluates how attention placement, number of heads, and attention dropout affect generalization. The main research question is: *Does adding spatial multi-head self-attention to a strong VGG-style ConvNet improve CIFAR-10 generalization?* The experimental evidence consists of baseline-building ablations, a 63-configuration attention grid search, and a repeated five-seed comparison. All model selection used stratified validation accuracy; the test set was reserved for final comparison. The best attention model used attention after all three VGG blocks with four heads and no attention dropout. It reached 89.65% mean test accuracy, compared with 89.26% for the no-attention baseline. In final 49k/1k single-run experiments, the same attention model reached 90.02% test accuracy, compared with 89.74% for the baseline. The conclusion is that multi-head spatial self-attention can provide a small but consistent improvement, but the gain is modest and not automatic across attention configurations.

1 Introduction

Convolutional neural networks (CNNs) remain a strong baseline for image classification because local filters, weight sharing, and progressive downsampling match the spatial structure of natural images. However, standard convolution is local: a feature at one spatial position is computed from nearby positions, and long-range interactions are only accumulated gradually as the receptive field grows. Attention mechanisms provide a complementary operation by allowing each spatial position to directly aggregate information from other positions. This motivates the question studied in this project: can a small CNN trained from scratch on CIFAR-10 benefit from explicitly adding spatial self-attention modules?

The task is 10-class image classification on CIFAR-10 (Krizhevsky, 2009). The project follows Default Project 2: start from the Assignment 3 patchify ConvNet, build a stronger ConvNet with VGG-style blocks, add regularization and a modern training recipe, and then investigate an extension based on attention modules. The main research question is:

Does adding spatial multi-head self-attention to a strong VGG-style ConvNet improve CIFAR-10 generalization?

The report contributes three pieces of evidence: a strong no-attention baseline, ablations showing which baseline components mattered, and a controlled attention study over placement, head count,

and dropout. The final result is positive but limited in size: the best attention model improves repeated mean test accuracy from 89.26% to 89.65% and final single-run test accuracy from 89.74% to 90.02%. The gain is therefore interpreted as a modest architectural improvement rather than a dramatic change in performance.

2 Related work

The baseline architecture is inspired by VGG networks, which showed that stacking multiple small 3×3 convolutions with increasing channel depth can produce strong image representations (Simonyan and Zisserman, 2015). VGG-style blocks are a natural extension of the Assignment 3 patchify ConvNet because they preserve a simple convolutional design while increasing depth, nonlinearity, and representational capacity.

Attention has become a central component in modern deep learning. Multi-head self-attention was popularized by the Transformer, where token representations are updated by content-dependent weighted combinations of other tokens (Vaswani et al., 2017). For images, attention can be used as the main modeling primitive, as in Vision Transformers (Dosovitskiy et al., 2021), or as an auxiliary module inside CNNs. Several CNN attention mechanisms have been proposed, including squeeze-and-excitation blocks that reweight channels (Hu et al., 2018), CBAM modules that combine channel and spatial attention (Woo et al., 2018), and non-local blocks that compute long-range interactions over space-time feature maps (Wang et al., 2018). This project focuses on a simple spatial self-attention module inserted into an otherwise standard CNN feature extractor.

The training recipe also uses common modern components: AdamW for decoupled weight decay (Loshchilov and Hutter, 2019), warmup followed by cosine decay related to cosine annealing schedules (Loshchilov and Hutter, 2017), and random erasing as a cutout-style data augmentation (Zhong et al., 2020).

3 Data and experimental protocol

CIFAR-10 contains 50,000 training images and 10,000 test images. Each image is an RGB image of size 32×32 , and the labels belong to ten object classes. The data were normalized using the standard CIFAR-10 channel statistics

$$\mu = (0.4914, 0.4822, 0.4465), \quad \sigma = (0.2470, 0.2435, 0.2616).$$

Training augmentation consisted of random horizontal flips, random crops with four-pixel padding, color jitter, and random erasing. Validation and test images used only tensor conversion and normalization.

Classification accuracy is the main evaluation metric. All model-selection experiments used a stratified 45k/5k train/validation split. This includes the baseline choices, the attention grid search, and the repeated comparison between candidate attention configurations. Validation accuracy was the selection metric throughout. The CIFAR-10 test set was held out from model selection and was used only for the final accuracy comparison between the selected no-attention baseline and the selected attention model. After this selection step, a 49k/1k train/validation version of the two final configurations was also run to obtain final learning curves while training on slightly more data.

CIFAR-10 is a heavily studied and relatively saturated benchmark, and state-of-the-art systems can exceed the accuracies reported here by a large margin. These published high-performing systems are therefore best treated as context rather than as direct competitors: they often use different model scales, augmentation recipes, training lengths, and hyperparameter budgets. The fair quantitative comparison in this report is therefore against the strong internal no-attention baseline and against different versions of the attention-augmented model.

4 Methods

4.1 VGG-style ConvNet baseline

The baseline starts with a patchify convolution that downsamples the input image using a 2×2 convolution with stride 2 and 64 output channels. This is followed by three VGG-style blocks

with channel widths [64, 128, 256]. Each block applies two serial 3×3 convolutional layers. Each convolution is followed by batch normalization and ReLU, with optional Dropout2d used in some regularization experiments. The first two blocks end with 2×2 max pooling, while the final block skips pooling to avoid excessive downsampling. The classifier uses global average pooling followed by a linear layer from 256 features to 10 class logits. The final no-attention baseline has 1,185,866 trainable parameters.

The best baseline recipe used AdamW, weight decay 10^{-3} , learning rate $2 \cdot 10^{-3}$, batch size 256, 100 epochs, five epochs of linear warmup, and cosine decay afterwards. The baseline design was intentionally kept simple and convolutional so that the extension isolated the effect of spatial self-attention. Alternatives such as label smoothing, convolutional dropout, and strided-convolution downsampling were tested during baseline development but were not kept when they did not improve validation accuracy.

4.2 Spatial multi-head self-attention

The attention module operates on a CNN feature map $X \in \mathbb{R}^{B \times C \times H \times W}$. The spatial dimensions are flattened into tokens,

$$T = \text{flatten}(X) \in \mathbb{R}^{B \times HW \times C},$$

so each spatial location is treated as one token with a C -dimensional feature vector. A pre-normalized multi-head self-attention block then computes

$$T' = T + \text{MHA}(\text{LayerNorm}(T)),$$

and T' is reshaped back to $B \times C \times H \times W$. This keeps the feature-map shape unchanged and makes the attention module a residual plug-in after any VGG block.

Attention was evaluated as an architectural addition rather than as a replacement for convolution. The convolutional blocks still learn local features and perform downsampling, while the attention module gives each spatial token a direct way to mix information with other spatial positions at the same feature scale. The selected attention model inserts this module after all three VGG blocks, uses four attention heads, and uses no attention dropout. It has 1,532,618 trainable parameters, about 29% more than the no-attention baseline, so the result should be interpreted as an architectural comparison rather than a parameter-matched efficiency claim.

5 Experiments and results

5.1 Building the strong no-attention baseline

The baseline was built by starting from the Assignment 3 patchify ConvNet and adding one change at a time. Table 1 reports validation accuracy, since these experiments were part of model selection. The test set is intentionally not used in this table. The table should therefore be read as a representative ablation study of the baseline recipe, not as final test-set reporting. In addition to the listed architectural and regularization ablations, we also performed moderately extensive hyperparameter tuning over settings such as learning rate, batch size, dropout, weight decay, augmentation choices, and training length. These details are omitted for brevity, and the table focuses on the main changes that most directly shaped the final no-attention baseline.

Table 1: Validation accuracy during baseline development on the stratified 45k/5k train/validation split. Section rows indicate whether the experiment changes depth, regularization, or the final three-block design. Values are percentages.

Model variant	Final val. acc.
<i>Reference and depth experiments</i>	
Assignment 3 patchify ConvNet	58.10
VGG-style ConvNet, 1 block, no extra regularization	66.59
VGG-style ConvNet, 3 blocks, no extra regularization	73.92
<i>Regularization ablations on the 3-block VGG-style ConvNet</i>	
3 blocks + weight decay only	70.63
3 blocks + FC-layer dropout only	79.36
3 blocks + data augmentation only	82.04
3 blocks + data augmentation + FC-layer dropout + weight decay	89.15
<i>Refinements of the regularized 3-block VGG-style ConvNet</i>	
3 blocks + data aug. + FC-layer dropout + weight decay + warmup-cosine LR	89.65
3 blocks + data aug. + FC-layer dropout + weight decay + warmup-cosine LR + strided downsampling	88.37
3 blocks + data aug. + Conv-layers dropout + weight decay + warmup-cosine LR + global average pooling	89.60
<i>Selected no-attention baseline</i>	
3 blocks + data augmentation + weight decay + warmup-cosine LR + global average pooling	90.16

The progression also shows why a strong baseline is important for the attention extension. A weak baseline would have made almost any added module look useful. Here, the no-attention model already used a modern training recipe and reached about 90% validation accuracy, so the attention module had to improve on a competitive point of comparison.

Several negative results were useful for model design. Replacing max pooling with strided convolutions reduced validation accuracy in the final recipe, suggesting that explicit pooling was still a good inductive bias for this small VGG-style network. Label smoothing and dropout also became less useful once global average pooling and strong augmentation were used. This is plausible because the classifier head was already small, random erasing and color jitter regularized the input space, and AdamW supplied weight decay throughout training.

5.2 Attention grid search

The attention grid search varied three design choices. First, the attention module could be inserted after the first, second, or third VGG block; every non-empty placement pattern was tested, from one attended block to all three attended blocks. Second, the number of attention heads was set to 2, 4, or 8. Third, attention dropout was set to 0.0, 0.05, or 0.1. This gives $7 \times 3 \times 3 = 63$ attention configurations. All grid-search runs used the same 45k/5k stratified split and were ranked by validation accuracy after 60 training epochs. Table 2 shows the strongest configurations that were carried forward.

Table 2: Best attention configurations from the 60-epoch grid search. Block numbers are one-based and refer to insertion after the corresponding VGG block. Values are percentages.

Attention configuration	Final val. acc.	Final val. loss
After blocks 2–3, 4 heads, dropout 0.1	90.08	0.3519
After blocks 1 and 3, 2 heads, dropout 0.0	89.90	0.3636
After block 2, 2 heads, dropout 0.0	89.80	0.3494
After blocks 1–3, 4 heads, dropout 0.0	89.80	0.3643
Known no-attention baseline after 100 epochs	90.16	0.4062

The 60-epoch grid search was used only as a cheaper screening stage for finding promising attention configurations. The no-attention baseline is included as a reference from the already established 100-epoch baseline, but it is not the main comparison in this table. The relevant result is therefore the ranking among attention variants. These candidates were then evaluated properly in the 100-epoch repeated-seed comparison against the no-attention baseline.

5.3 Repeated comparison against the baseline

The top four attention configurations and the no-attention baseline were retrained for 100 epochs over five seeds: 26, 27, 28, 29, and 30. The attention model was selected by mean final validation accuracy. Test accuracy was computed after training to compare the selected attention model with the baseline, but it was not used to choose the configuration. Results are shown in Table 3.

Table 3: Repeated 5-seed comparison on the 45k/5k stratified split. The best attention model was selected by mean final validation accuracy; test accuracy was reported only after this validation-based selection and was not used for model selection. The standard deviation is computed from the reported variance. Values are percentages.

Configuration	Mean final val.	Mean test $\pm$ sd
After blocks 1–3, 4 heads, dropout 0.0	90.51	89.65 $\pm$ 0.16
After blocks 1 and 3, 2 heads, dropout 0.0	90.37	89.63 $\pm$ 0.30
After blocks 2–3, 4 heads, dropout 0.1	90.15	89.60 $\pm$ 0.15
After block 2, 2 heads, dropout 0.0	90.14	89.19 $\pm$ 0.19
No-attention baseline	90.07	89.26 $\pm$ 0.23

The best attention model improved mean final validation accuracy by 0.44 percentage points and mean test accuracy by 0.39 percentage points over the baseline. The gain is small relative to the run-to-run variation, but it appeared in both validation and test accuracy and was reproduced in the final 49k/1k run. The most stable conclusion is therefore not that attention is a large improvement, but that spatial self-attention can add a small amount of useful global mixing on top of an already strong convolutional feature extractor.

The placement pattern also matters. Attention after all three blocks was better than the best late-block-only model in the repeated comparison, even though it was not the top model in the shorter grid search. A likely explanation is that the module is useful at several spatial scales: early feature maps retain more spatial detail, while later feature maps contain more semantic channel structure. Four heads were sufficient; increasing to eight heads did not reliably improve validation accuracy, possibly because CIFAR-10 images are small and the feature maps after downsampling already contain relatively few spatial tokens.

5.4 Final 49k/1k runs

After selecting the final configurations, both the baseline and best attention model were trained once using 49,000 training examples and a 1,000-image stratified validation set. These runs were used for final learning curves and final test-set reporting. The no-attention baseline reached 89.74% test accuracy. The attention model reached 90.02% test accuracy, as shown in Table 4.

Table 4: Final 49k/1k single-run comparison. The attention model achieves the best test accuracy, but the effect remains modest.

Model	Train acc.	Test acc.
No-attention baseline	99.60	89.74
Attention after all three blocks	99.64	90.02

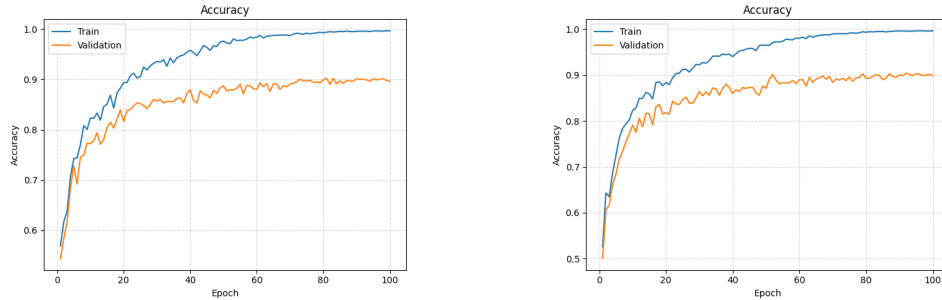


Figure 1: Final accuracy curves for the selected no-attention baseline (left) and the selected attention model (right). Both models closely fit the training data, so the relevant difference is the small test improvement from attention rather than training accuracy.

Both final models fit the training set almost completely. This means that the limiting factor is not optimization capacity but generalization. The baseline generalizes well because augmentation and weight decay constrain the model while global average pooling prevents the classifier from becoming too large. The attention model then adds a different inductive bias: it allows spatial positions to exchange information directly after each convolutional stage. On CIFAR-10 this benefit is naturally limited, because the images are small and the convolutional receptive field already grows quickly. This explains why the improvement is consistent but modest, and why attention dropout was not necessary in the final selected configuration: the rest of the recipe already regularized the model strongly. A limitation of the analysis is that it uses aggregate accuracy and learning curves rather than a per-class confusion matrix; thus the report can support a small average improvement, but not a claim that attention fixes a specific failure mode such as confusing visually similar classes.

6 Conclusion

The most important baseline improvements were VGG-style depth, data augmentation, AdamW weight decay, warmup plus cosine learning-rate decay, and global average pooling. The extension then evaluated spatial multi-head self-attention modules inserted into CNN feature maps.

The answer to the research question is cautiously positive. The best attention configuration, using attention after all three VGG blocks with four heads and no attention dropout, improved mean test accuracy from 89.26% to 89.65% in a five-seed comparison and improved final single-run test accuracy from 89.74% to 90.02%. However, the benefit depended on the attention configuration, and the final gain was less than half a percentage point. The main lesson is therefore not that attention automatically improves CNNs, but that carefully placed spatial self-attention can provide a modest generalization benefit when added to an already strong ConvNet.

Future work should test whether the improvement remains under stricter controls, for example by comparing against a slightly wider no-attention CNN with a similar parameter count and training time. It would also be useful to add per-class error analysis, example-level failure cases, and calibration metrics to understand whether attention helps specific CIFAR-10 classes or mainly improves average accuracy through small gains across many classes.

GitHub Repository

<https://github.com/JonesMea/cnn-spatial-self-attention-cifar10>

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